

Project Documentation

Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Introduction

• Project Title:

Visualization Tool for Electric Vehicle Charge and Range Analysis

2. Project Overview

Purpose:

The objective of this project is to design and implement an interactive visualization platform for analyzing Electric Vehicle (EV) charging infrastructure, performance metrics, brand distribution, powertrain segmentation, and pricing trends.

The project integrates MySQL for data preprocessing, Tableau Desktop for interactive dashboard development, and a responsive HTML/CSS webpage to provide an accessible analytics solution.

Features:

- Interactive EV analytics dashboard
- Charging station distribution analysis
- Brand and model comparison
- Performance and efficiency analysis
- Powertrain segmentation
- Price trend visualization
- Dashboard publishing using Tableau Public
- Responsive HTML/CSS interface for accessing the visualization.

2. System Architecture

Frontend:

The frontend is developed using HTML and CSS to provide a simple and responsive interface.

The webpage contains navigation links and provides access to the published Tableau Public dashboard and story.

Database:

The project uses MySQL for:

Data cleaning

Null value removal

Data aggregation

Data standardization

Data preprocessing before visualization

The processed data is then imported into Tableau Desktop for creating visualizations.

Architecture Flow:

Raw EV datasets (CSV files) are collected.

Data preprocessing and cleaning are performed using MySQL.

Cleaned data is imported into Tableau Desktop.

Interactive dashboards and stories are created.

Dashboards and stories are published to Tableau Public.

The Tableau Public visualization is accessed through a simple HTML/CSS webpage.

Users can view the dashboard and story through their web browser.

3.Setup Instructions:

Prerequisites

- MySQL
- Tableau Desktop
- Web Browser (Chrome or Edge)
- Internet connection (for Tableau Public)

• Dashboard and Story pdf/

Contains exported PDF versions of the Tableau Dashboard and Story for submission and offline reference.

• Datasets/

Contains all CSV datasets used for data cleaning, preprocessing, and visualization development.

• Documentation/

Contains complete project documentation including Ideation Phase, Requirement Analysis, Design, Planning, Development, Testing, and Final Documentation.

• Screenshots/

Contains dashboard screenshots, analysis visuals, and web interface images used for reporting and demonstration.

•HTML/

Contains the HTML and CSS files used to create the project webpage.

4.Running the Application

To view the project locally:

Step 1: Open the project folder.

Step 2: Open the **index.html** file in any web browser.

Step 3: Use the navigation menu to access the Tableau Public dashboard or story.

Alternatively, the dashboard can be viewed directly through its Tableau Public link.

5. API Documentation

This project does not expose any REST APIs.

Data preprocessing is performed using MySQL queries, while visualization is handled using Tableau Desktop and Tableau Public.

6. Authentication

The Tableau dashboard is hosted on Tableau Public and is publicly accessible.

No authentication or user login system is implemented in this project.

7. User Interface:

The user interface is developed using **HTML and CSS**.

- It includes:
- Home page
- Navigation bar
- Project introduction
- Dashboard access
- Story access
- About section
- Contact section
- Screenshots of the webpage, dashboard, and story are included in the documentation.

8. Testing

Testing was performed through:

- Validation of MySQL preprocessing queries
- Verification of cleaned data
- Dashboard functionality testing in Tableau Desktop
- Verification of filters and interactive visualizations
- Story navigation testing
- Cross-browser compatibility testing (Google Chrome and Microsoft Edge)
- Testing of HTML webpage navigation

Verification of Tableau Public accessibility

